DRUGS · ANTIBIOTICS: PENICILLINS

# Amoxicillin

Amoxicillin is a broad-spectrum aminopenicillin and the antibiotic an ENT doctor prescribes most often: first-line for acute otitis media in children, a choice in acute bacterial sinusitis, and the base on which co-amoxiclav is built. It is cheap, well absorbed and safe, with two traps: it is inactivated by beta-lactamase, and in a patient with glandular fever it causes a widespread rash.

## 1 CHAPTER ONE At a glance

The drug on one page.

### 1.1 Summary

AMOXICILLIN

| | SUMMARY |
|---|---|
| **Class** | Aminopenicillin (beta-lactam) |
| **How it works** | Binds penicillin-binding proteins and blocks cross-linking of the bacterial cell wall; bactericidal |
| **Spectrum** | Streptococcus pneumoniae, Streptococcus pyogenes, non-beta-lactamase Haemophilus influenzae and Moraxella, enterococci; not Staphylococcus aureus (most produce penicillinase), not Pseudomonas, not anaerobes of the beta-lactamase type |
| **ENT uses** | Acute otitis media; acute bacterial sinusitis; early Lyme disease; dental and minor soft tissue infection; endocarditis prophylaxis in selected cases; Helicobacter eradication |
| **Adult dose** | 500 mg three times daily by mouth; 1 g three times daily for severe infection |
| **Child dose** | 1 to 11 months 125 mg, 1 to 4 years 250 mg, 5 to 17 years 500 mg, each three times daily (or 30 mg/kg up to 1 g three times daily for severe infection) |
| **Duration** | 5 to 7 days for acute otitis media; 5 days for sinusitis |
| **Avoid** | Penicillin allergy; suspected infectious mononucleosis |

| | SUMMARY |
|---|---|
| **Pregnancy** | Safe; compatible with breastfeeding |

# 2 CHAPTER TWO How it works

Mechanism and handling.

## 2.1 Mechanism

Amoxicillin inhibits the transpeptidases (penicillin-binding proteins) that cross-link peptidoglycan, so dividing bacteria lyse. Killing is **time-dependent**: what matters is how long the free drug stays above the minimum inhibitory concentration, which is why it is given three times a day and why higher doses overcome intermediate pneumococcal resistance. Resistance comes from **beta-lactamase** (about a fifth to a third of H. influenzae and almost all Moraxella catarrhalis) and from altered penicillin-binding proteins in pneumococci.

## 2.2 Absorption and handling

Oral absorption is about 90% and is not reduced by food, unlike ampicillin. It is excreted unchanged by the kidney; reduce the dose when the eGFR is below 30 ml/min. It penetrates middle ear fluid and the sinuses well but crosses the uninflamed blood-brain barrier poorly.

# 3 CHAPTER THREE Using it in ENT

Indications and doses.

## 3.1 Acute otitis media

Most acute otitis media resolves without antibiotics, so NICE advises **no antibiotic or a delayed (back-up) prescription** for most children, with regular paracetamol or

ibuprofen. Give an immediate antibiotic to children who are systemically unwell or at high risk of complications, those under 2 with bilateral disease, and children with otorrhoea from a perforation. Amoxicillin is first-line for **5 to 7 days**; clarithromycin or erythromycin if allergic; co-amoxiclav as second-line when symptoms do not improve after 2 to 3 days.

### 3.2 Acute sinusitis

Symptoms for under 10 days need no antibiotic. For symptoms lasting more than 10 days, a back-up prescription can be considered; phenoxymethylpenicillin is the NICE first choice in adults, and amoxicillin is widely used elsewhere and in children. Co-amoxiclav is kept for patients who are systemically very unwell or have signs of a complication, who need assessment rather than a prescription.

### 3.3 Other uses

- **Lyme disease** (erythema migrans, facial palsy): amoxicillin or doxycycline, for 14 to 21 days.
- **Dental infection** with spreading cellulitis: often with metronidazole.
- **Helicobacter pylori**: as part of triple therapy.

## 4 CHAPTER FOUR Safety

What goes wrong.

### 4.1 Adverse effects

- **Diarrhoea**, nausea and candidiasis; Clostridioides difficile infection is less common than with co-amoxiclav or cephalosporins.
- **Rash.** A delayed, itchy maculopapular rash appears in a few per cent of patients, often not true allergy. In **infectious mononucleosis** (and with cytomegalovirus and lymphatic leukaemia) a widespread maculopapular rash is common; it is not IgE-mediated and does not prove penicillin allergy, but it frightens patients and clinicians into lifelong labels.

- **True hypersensitivity**: urticaria, angioedema and anaphylaxis (about 1 in 5,000 to 10,000 courses).
- Rarely interstitial nephritis, seizures at high doses in renal failure, and haemolytic anaemia.

### 4.2 Contraindications and cautions

- **Penicillin allergy** of any type (immediate or severe delayed).
- **Glandular fever** suspected: avoid; use phenoxymethylpenicillin or clarithromycin for a sore throat with exudate when an antibiotic is truly needed.
- Renal impairment: dose adjustment.

### 4.3 Interactions

Increases the effect of **warfarin** (check the INR). Reduces the excretion of **methotrexate**. Allopurinol increases the risk of rash.

## 5 CHAPTER FIVE For the examination

What the examiner wants to hear.

### 5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- Spectrum and why it fails against **beta-lactamase producers**.
- NICE approach to **acute otitis media**: no or back-up antibiotic for most; who needs one at once; 5 to 7 days.
- Why amoxicillin is avoided in a sore throat that may be **glandular fever**.
- Time-dependent killing and dosing frequency.

## 5.2 Common pitfalls

COMMON PITFALLS

- Giving amoxicillin for an exudative tonsillitis in a teenager without considering mononucleosis.
- Labelling a mononucleosis rash as lifelong penicillin allergy.
- Prescribing amoxicillin for an otitis externa or a perichondritis (staphylococci and Pseudomonas).
- Prescribing for a viral upper respiratory infection.

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